

D420

Discrete Math: Logic

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A software tester reviewing a payroll module reads the rule “If an employee is salaried, then overtime is not paid.” The module was written in Java by an outside contractor. In which case is the rule false?

- A. An hourly employee is not paid overtime.
- B. An hourly employee is paid overtime.
- C. It depends on the programming language used.
- D. A salaried employee is paid overtime.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

The statement $p \rightarrow q$ is known to be false. What are the truth values of p and q?

- A. It cannot be determined.
- B. Both are false.
- C. p is true and q is false.
- D. p is false and q is true.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A network policy says: “If a device is compliant, then it gets network access.” The policy was drafted by a committee of six and approved in March. Which statement is logically equivalent?

- A. If a device is not compliant, then it does not get network access.
- B. If a device gets network access, then it is compliant.
- C. A device gets network access if and only if it is compliant.
- D. If a device does not get network access, then it is not compliant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

What is the converse of “If n is divisible by 4, then n is even,” and is it true for all integers?

- A. “If n is not even, then n is not divisible by 4”; true.
- B. “If n is even, then n is divisible by 4”; false ($n = 6$).
- C. “If n is even, then n is divisible by 4”; true, since a converse always matches the original.
- D. “If n is not divisible by 4, then n is not even”; false.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A firewall rule should match traffic that is NOT (from the trusted subnet AND on port 443). The rule set was migrated from an older vendor last spring. Which condition is equivalent?

- A. From the trusted subnet OR on port 443.
- B. Not from the trusted subnet AND not on port 443.
- C. Not from the trusted subnet OR not on port 443.
- D. From the trusted subnet AND not on port 443.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A QA lead claims, “Every test in the suite passed.” The suite has 1,340 tests and runs on three build servers. What is the correct negation of the claim?

- A. At least one test in the suite did not pass.
- B. At least one test in the suite passed.
- C. No test in the suite passed.
- D. At least one of the three build servers failed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Over the integers, what is the negation of $\exists x (x > 5)$?

- A. $\forall x (x \leq 5)$
- B. $\exists x (x \leq 5)$
- C. $\forall x (x < 5)$
- D. $\neg \forall x (x > 5)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An analyst reasons: "If the server is overloaded, response time goes over 2 seconds. Response time went over 2 seconds this morning. So the server was overloaded." The server has 64 GB of RAM. How should the argument be judged?

- A. Valid by modus tollens.
- B. Valid, because 64 GB of RAM cannot handle the load.
- C. Valid by modus ponens.
- D. Invalid: it affirms the consequent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A student wants to prove: "For every integer n , if n^2 is even, then n is even." She has a strong algebra background and a week to finish. Which approach is the cleanest?

- A. Assume n^2 is odd and show n is odd.
- B. Check $n = 1$ through 10 and note the pattern holds.
- C. Contrapositive: assume n is odd and show n^2 is odd.
- D. Find a counterexample.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A logic designer has the output $F = A + A \cdot B$ for a door-alarm circuit. The circuit uses 5-volt TTL parts from existing stock. What is the simplest form of F ?

- A. $F = A + B$
- B. $F = A$
- C. $F = B$
- D. $F = A \cdot B$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A designer builds a truth table for a 4-input function in a vending machine controller; the machine holds 30 products. How many rows does the truth table have?

- A. 16
- B. 65,536
- C. 30
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

How many distinct Boolean functions of 3 variables exist?

- A.** 3
- B.** 256
- C.** 8
- D.** 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A software tester reviewing a payroll module reads the rule “If an employee is salaried, then overtime is not paid.” ~~The module was written in Java by an outside contractor.~~ In which case is the rule false?

Target word: case where $p \rightarrow q$ is false **Decides:** Hypothesis true (salaried) and conclusion false (overtime paid).

A	Confusion set	False hypothesis, so the conditional is true.
B	Inverted operation	Swaps the roles of hypothesis and conclusion; a false hypothesis makes the rule true.
C	Camouflage	The truth table does not depend on how the module was built.
D	Correct	True hypothesis with a false conclusion is the only false case.

2. Answer C Concept 1 • transfer

The statement $p \rightarrow q$ is known to be false. What are the truth values of p and q ?

Target word: solve for inputs from the output **Decides:** Only one row of the table is false.

A	Dropped element	Only one row is false, so both values are fixed.
B	Confusion set	A false hypothesis makes the conditional true.
C	Correct	That is the single false row.
D	Inverted operation	That row is true.

3. Answer D Concept 2

A network policy says: “If a device is compliant, then it gets network access.” ~~The policy was drafted by a committee of six and approved in March.~~ Which statement is logically equivalent?

Target word: equivalent form **Decides:** Only the contrapositive is equivalent.

A	Confusion set	The inverse; not equivalent.
B	Inverted operation	The converse; not equivalent.
C	Adjacent category	A biconditional claims more than the original.
D	Correct	The contrapositive $\neg q \rightarrow \neg p$.

4. Answer B Concept 2 • transfer

What is the converse of “If n is divisible by 4, then n is even,” and is it true for all integers?

Target word: form and truth of the converse **Decides:** 6 is even but not divisible by 4.

A	Confusion set	That is the contrapositive, not the converse.
B	Correct	Converse swaps hypothesis and conclusion; 6 is a counterexample.
C	Absolute language	Converses do not share the original’s truth value in general.
D	Adjacent category	That is the inverse.

5. Answer C Concept 3

A firewall rule should match traffic that is NOT (from the trusted subnet AND on port 443). ~~The rule set was migrated from an older vendor last spring.~~ Which condition is equivalent?

Target word: apply De Morgan **Decides:** Negate each part and flip AND to OR.

A	Inverted operation	Flips the connective but drops the negations.
B	Dropped element	Negates both parts but forgets to flip AND to OR.
C	Correct	$\neg(p \wedge q) \equiv \neg p \vee \neg q$.
D	Confusion set	Only one case of the correct OR.

6. Answer A Concept 4

A QA lead claims, "Every test in the suite passed." ~~The suite has 1,340 tests and runs on three build servers.~~ What is the correct negation of the claim?

Target word: negate a universal **Decides:** $\neg\forall$ becomes $\exists\neg$.

A	Correct	$\neg\forall x P(x) \equiv \exists x \neg P(x)$.
B	Inverted operation	Compatible with the original claim, so not its negation.
C	Absolute language	Far stronger than the negation.
D	Camouflage	The servers are not part of the claim.

7. Answer A Concept 4 · transfer

Over the integers, what is the negation of $\exists x (x > 5)$?

Target word: negate an existential **Decides:** $\neg\exists$ becomes $\forall\neg$, and $\neg(x > 5)$ is $x \leq 5$.

A	Correct	Flip the quantifier and negate the predicate.
B	Dropped element	Negates the predicate but keeps the quantifier.
C	Confusion set	The negation of $>$ is $\leq$, not $<$.
D	Inverted operation	Equals $\exists x (x \leq 5)$, not the negation.

8. Answer D Concept 5

An analyst reasons: "If the server is overloaded, response time goes over 2 seconds. Response time went over 2 seconds this morning. So the server was overloaded." ~~The server has 64 GB of RAM.~~ How should the argument be judged?

Target word: validity of the form **Decides:** Reasons from q back to p.

A	Adjacent category	Modus tollens needs $\neg q$ to conclude $\neg p$.
B	Camouflage	Memory size does not change the argument's form.
C	Confusion set	Modus ponens starts from p, not q.
D	Correct	$p \rightarrow q$ and q do not guarantee p; other causes could slow the response.

9. Answer C Concept 6

A student wants to prove: "For every integer n , if n^2 is even, then n is even." ~~She has a strong algebra background and a week to finish.~~ Which approach is the cleanest?

Target word: best proof method **Decides:** Starting from n odd is easy to work with.

A	Inverted operation	That proves the converse's contrapositive, a different statement.
B	Dropped element	Examples skip the infinitely many other integers.
C	Correct	$n = 2k + 1$ gives $n^2 = 2(2k^2 + 2k) + 1$, which is odd.
D	Adjacent category	Counterexamples disprove claims; this one is true.

10. Answer B Concept 7

A logic designer has the output $F = A + A \cdot B$ for a door-alarm circuit. ~~The circuit uses 5-volt TTL parts from existing stock.~~ What is the simplest form of F ?

Target word: simplify **Decides:** Absorption: $A + AB = A$.

A	Confusion set	That is the result of $A + A \cdot B$, not $A + AB$.
B	Correct	$A(1 + B) = A \cdot 1 = A$.
C	Dropped element	Drops the A term that dominates the expression.
D	Inverted operation	Treats the OR as an AND.

11. Answer A Concept 8

A designer builds a truth table for a 4-input function in a vending machine controller; ~~the machine holds 30 products.~~ How many rows does the truth table have?

Target word: rows for 4 inputs **Decides:** Rows = 2^n .

A	Correct	$2^4 = 16$.
B	Adjacent category	That is the number of distinct functions, 2^{16} .
C	Camouflage	The product count does not set the number of inputs.
D	Inverted operation	2×4 multiplies instead of raising to a power.

12. Answer B Concept 8 · transfer

How many distinct Boolean functions of 3 variables exist?

Target word: solve for functions, not rows **Decides:** $2^{(2^3)} = 2^8$.

A	Wrong stem number	The number of variables.
B	Correct	8 rows, each output 0 or 1: $2^8 = 256$.
C	Confusion set	That is the number of rows.
D	Inverted operation	2×3 instead of exponentiation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Truth of a conditional	$p \rightarrow q$ is false only when p is true and q is false; a false hypothesis makes it true.	Thinking a false hypothesis makes the conditional false.	"If ... then ..." and "when is it false?"
2	Converse, inverse, contrapositive	Only the contrapositive ($\neg q \rightarrow \neg p$) is equivalent to $p \rightarrow q$; the converse and inverse are equivalent to each other, not to the original.	Treating the converse as equivalent.	"Logically equivalent" with an if-then statement.
3	De Morgan's laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$; negate each part and flip the connective.	Negating both parts but keeping AND or OR.	"NOT (A and B)."
4	Negating quantifiers	$\neg \forall x P(x) \equiv \exists x \neg P(x)$, and $\neg \exists x P(x) \equiv \forall x \neg P(x)$.	Negating "every" as "none."	"All," "every," "some," "there exists," then "negation."
5	Valid argument forms	Modus ponens ($p \rightarrow q, p \vdash q$) and modus tollens ($p \rightarrow q, \neg q \vdash \neg p$) are valid; affirming the consequent and denying the antecedent are fallacies.	Reasoning from q back to p .	An if-then premise followed by the "then" part observed.
6	Choosing a proof method	Direct proof assumes p to show q ; contrapositive assumes $\neg q$ to show $\neg p$; contradiction assumes the claim false; examples cannot prove a universal claim, but one counterexample disproves it.	Proving a universal claim by checking examples.	"For every integer ..." with a hypothesis that is hard to use directly.
7	Boolean simplification	Absorption: $A + AB = A$ and $A(A + B) = A$; also $A + A'B = A + B$.	Mixing up absorption with $A + A'B$.	An expression where a variable appears alone and inside a product.
8	Counting truth tables and functions	n inputs give 2^n truth-table rows and $2^{(2^n)}$ distinct Boolean functions.	Multiplying $2 \times n$, or confusing rows with functions.	"How many rows" or "how many functions" for n variables.

Course scope (public description): Discrete Math-Logic is designed to help students develop competence in the use of logic and proofs and Boolean Algebra and Boolean functions. Applied Probability and Statistics and Applied Algebra are prerequisites for this course.